

Introduction to RISC-V and Processor Design

Overview of RISC-V Architecture, RUPES
Simulator, and HDL-based Processor Design

RISC vs CISC Comparison

RISC (Reduced Instruction Set Computer):

- Simple, fixed-length instructions.
- Fewer addressing modes.
- Emphasizes hardware simplicity and speed.

CISC (Complex Instruction Set Computer):

- Larger and more complex instruction set.
- Variable-length instructions.
- Emphasizes software compactness and versatility.

How Processors Are Designed



Architecture Definition – Define the instruction set (ISA).



HDL Implementation – Write Verilog/VHDL/SystemVerilog code for datapath, control unit, and memory.



Simulation – Test logic functionality using simulators like ModelSim or Vivado.



Synthesis – Convert HDL into gate-level representation.



Fabrication – Manufacture using semiconductor processes.



Verification – Validate design through testing and benchmarking.

Role of HDL in Processor Design



Hardware Description Languages (HDLs) like Verilog or VHDL describe hardware behavior and structure.



They define how control units, ALUs, registers, and buses interact.



Used for both simulation and synthesis of digital circuits.



Enables hardware engineers to move from abstract design to physical implementation.

Verilog vs VHDL vs SystemVerilog



Verilog: Hardware-oriented, C-like syntax, widely used in industry.

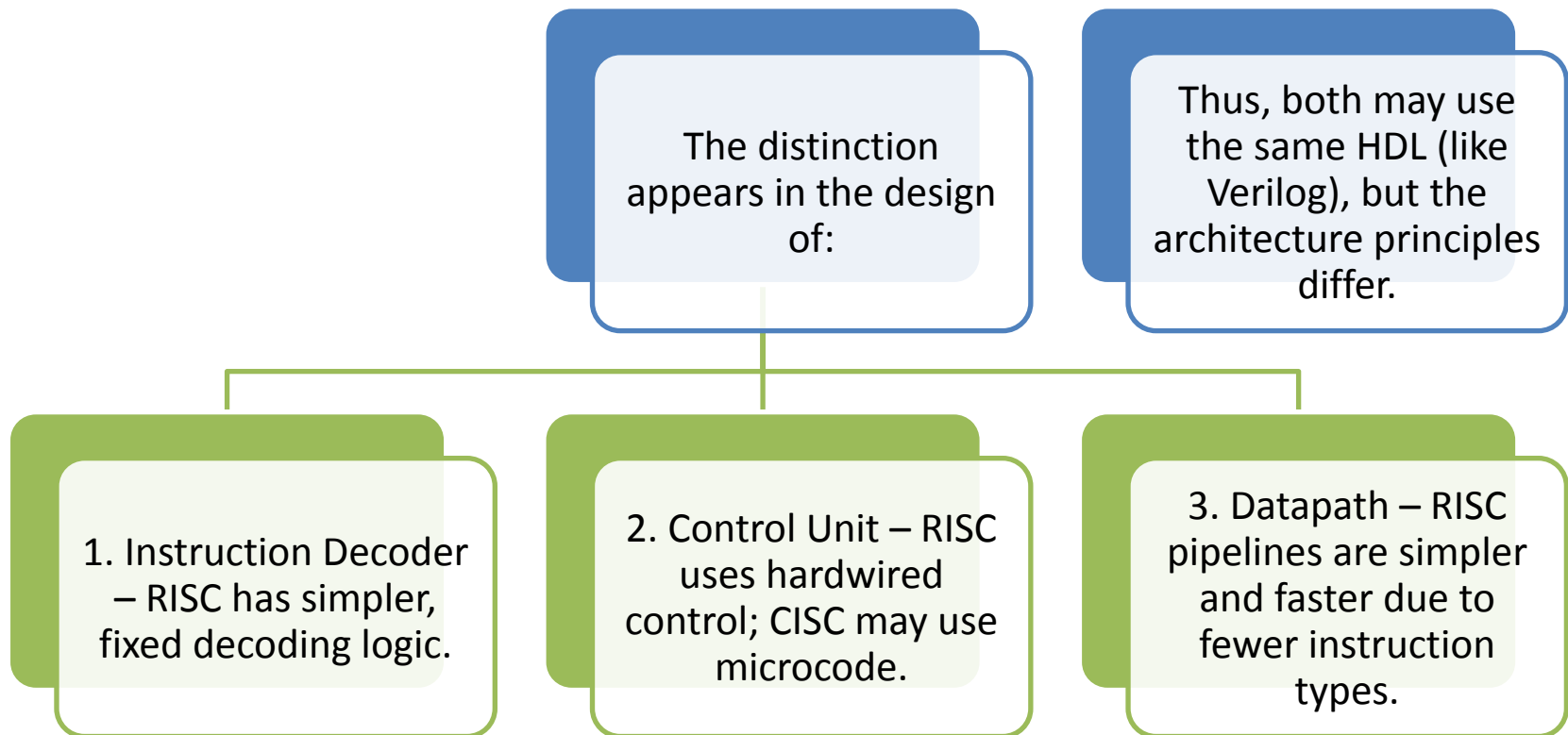


VHDL: Strongly typed, verbose, ADA-based syntax, often used in academia and defense.



SystemVerilog: Extension of Verilog with advanced features like object-oriented constructs, assertions, and testbench capabilities.

Where RISC vs CISC Differs in HDL



The Story of How a Processor Is Built

Imagine a team of engineers

Let's call them the **Silicon Architects**

They want to create a new processor, the beating heart of every computer.

Chapter 1: The Idea



It all starts with an idea — a vision of what the processor should do.



Idea: “We want a CPU that’s fast, efficient, and can run 64-bit software.”



The architects begin by designing the **architecture**



Defining how many registers it has, what instructions it understands, and how it communicates with memory.



This becomes the **Instruction Set Architecture (ISA)** — like **x86**, **ARM**, or **RISC-V**. It’s the *language* the processor will speak.

Chapter 2: The Blueprint (Microarchitecture)



Next, microarchitects decide *how* the processor will execute those instructions.



E.g : “Let’s have a 5-stage pipeline: Fetch, Decode, Execute, Memory, Write-back.”



They plan how data will move through the **datapath**, how instructions are decoded, and how the **ALU** performs math operations.



This is the **blueprint** — not yet a physical circuit, but a logical plan of how the processor will behave.

Chapter 3: Writing the Hardware (HDL Stage)

- Now the engineers start writing **HDL code** usually in **Verilog**, **VHDL**, or **SystemVerilog**.
- It looks like programming, but it actually **describes hardware behavior**.
- This tiny piece of code represents part of the **ALU**
- a real circuit that will one day add and subtract numbers in hardware.

```
always @(*) begin
  case(op)
    4'b0000: result = A + B;
    4'b0001: result = A - B;
    default: result = 0;
  endcase
end
```

Chapter 4: Simulation & Verification

- Before building anything physical, the HDL design is **simulated**.
- The engineers test every corner case:
 - Does the ALU give the right result?
 - Does the pipeline handle branches correctly?
 - Does memory respond on time?
- They use professional simulators like **ModelSim**, **VCS**, or **QuestaSim**.
- Write millions of **test benches** to verify every logic path.
- This is like checking every neuron in a brain before it's born.

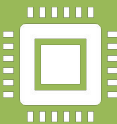
Chapter 5: Synthesis and Optimization



Once the design works perfectly, it goes through **synthesis** - the process of converting HDL code into **logic gates** (AND, OR, NOT).



The synthesis tool transforms abstract logic into a **gate-level netlist**, a map of how every tiny logic element connects.



Engineers then run **place and route** tools, which physically arrange these gates into layers of transistors that will fit inside the chip's tiny silicon die.

Chapter 6: The Silicon Birth (Fabrication)

Now comes the magical part — **tape-out**.

The final layout (the chip's physical map) is sent to a **semiconductor foundry** — like **TSMC**, **Intel Foundry**, or **Samsung**.

There, the design is printed onto silicon wafers using **photolithography**, layer by layer, atom by atom.

Millions — or even billions — of tiny **transistors** are etched and connected according to the HDL logic that started this whole story.

Each chip is then cut, packaged, and tested.

Chapter 7: Life Begins

At last, the processor is placed inside a computer.

It boots up, fetches its first instruction, and executes code — exactly as the architects imagined months or years before.

That simple Verilog line
`result = A + B;`

is now a physical circuit switching billions of times per second to perform real-world computation.

The Design Process: from HDL → Real Chip (Summary)

Stage	Description	Example
Architecture design	Engineers define the instruction set, registers, and features (e.g., x86-64, cache size, pipelining, etc.)	"We'll support Intel x86 + AVX instructions"
Microarchitecture design	Internal design — how instructions are executed, pipelines, ALUs, branch predictors, etc.	How many stages the pipeline will have
HDL implementation	Written in Verilog/SystemVerilog/VHDL	The logic of ALUs, decoders, register files, etc.
Simulation & Verification	Run tests (like Ripes does, but much larger scale)	Using simulators (ModelSim, VCS, QuestaSim)
Synthesis & Fabrication	HDL → Gate-level → Transistor-level → Physical chip	Using tools from Synopsys, Cadence, or Siemens
	Final layout sent to	

What is RISC-V



RISC-V is an open-source Instruction Set Architecture (ISA) based on the Reduced Instruction Set Computer (RISC) principle.



It is modular, extensible, and royalty-free.



Developed at UC Berkeley to promote education, research, and industrial use.



Supports 32-bit, 64-bit, and 128-bit implementations (RV32, RV64, RV128).

Industrial & Market Applications

Sector	How RISC-V Is Used	Example Companies / Products
IoT & Embedded Systems	Used in microcontrollers for smart sensors, wearables, and industrial control.	SiFive, Espressif ESP32-C3 (Wi-Fi/BLE MCU), GreenWaves GAP9
Consumer Electronics	Runs firmware and control processors in devices like SSDs, Wi-Fi modules, and smart TVs.	Western Digital (billions of RISC-V cores in HDD/SSD controllers), Alibaba T-Head
Automotive	Used for domain controllers, AI-assisted driving, and safety subsystems.	Bosch, Andes Technology, Semidynamics
Artificial Intelligence & ML	Custom RISC-V accelerators with vector or tensor extensions for efficient AI workloads.	Esperanto Technologies (ET-SoC-1, 1000+ RISC-V cores), Tenstorrent, Ventana

Operating Systems & Software Ecosystem

- RISC-V can **run modern OS and tools**, such as:
- **Linux, FreeBSD, Zephyr, RT-Thread**
- **Compilers:** GCC, LLVM/Clang
- **Emulators:** QEMU, Spike, TinyEMU
- **Debuggers & IDEs:** GDB, PlatformIO, VS Code extensions
- **Meaning:** You can write **C/C++**, **Python**, or **Rust** programs and compile them for a RISC-V target — just like you would for ARM or x86.



Strengths in the Market



Open and royalty-free: No licensing cost → fosters innovation.



Customizability: Companies can add instructions for domain-specific tasks (AI, DSP, encryption).



Ecosystem growth: Backed by the **RISC-V International consortium** (3,000+ members).



Education-friendly: Transparent, simple, and ideal for learning ISA and hardware-software co-design

Market Trend

- **Growing adoption:** Analysts project **>25 billion RISC-V cores shipped by 2030.**
- **China and Europe** are investing heavily in RISC-V for **technology independence.**
- **Major chipmakers** (Intel, Qualcomm, NVIDIA) are developing or supporting RISC-V toolchains.
- **Linux-capable RISC-V boards** (e.g., StarFive VisionFive 2, HiFive Unmatched) are now available for developers.



Real-World Examples

Product	Description
ESP32-C3	Wi-Fi/Bluetooth microcontroller powered by RISC-V core, popular in IoT.
HiFive Unmatched	A Linux-capable RISC-V PC board by SiFive.
Western Digital SweRV Core	Open-source RISC-V core used in storage controllers.
BeagleV® Ahead	BeagleBoard's RISC-V single-board computer for developers.
Ventana Veyron	High-performance data center-class RISC-V processors.

Top 4 Companies Using RISC-V

- **SiFive:** Founded by the original creators of RISC-V at UC Berkeley. Provides commercial RISC-V IP cores, SoCs, and design platforms. Acts as the leading industrial face of RISC-V.
- **Western Digital:** Replaced proprietary controllers with RISC-V cores in storage devices (SSDs and HDDs). Has shipped over **2 billion RISC-V cores**, making it one of the largest real-world adopters.
- **Alibaba (T-Head Semiconductor):** China's largest RISC-V contributor. Develops high-performance RISC-V processors (Xuantie series) used in AI, IoT, and edge devices.
- **Espressif Systems:** Maker of ESP32-C3 and ESP32-C6 chips, which use RISC-V cores. These chips dominate the **IoT and embedded systems** market worldwide due to low cost and open-source support.

Powering Innovation Across Industries

The SiFive® Core IP portfolio consists of four distinct families, each designed to address diverse computing demands: high-performance application processors, low-power embedded microcontrollers, highly scalable vector and matrix processors for AI and modern workloads, and solutions specifically designed for the automotive industry. Built on the RISC-V ISA and supported by flexible software and a global ecosystem, our core IP offers customers world-class performance and versatility.



Essential

The SiFive Essential™ family offers a wide range of silicon-proven, pre-defined embedded CPU cores or lets you use SiFive Core Designer to build a custom processor configuration tailored to your specific market and needs.

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- ✓ Highly customizable to meet exact requirements
- ✓ Scales from 32-bit MCUs to 64-bit dual-issue
- ✓ Proven Innovation: billions of units shipped worldwide



Intelligence

SiFive Intelligence™ embraces a software-first approach to processor design, addressing the future needs of deploying ML and AI at the edge. It provides high-performance, scalable vector and matrix compute capabilities, adaptable to evolving workloads.

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- ✓ Scalar + wide vector compute engines
- ✓ High bandwidth bus interfaces to control custom engines
- ✓ Highly scalable matrix computing solutions



Performance

SiFive Performance™ delivers maximum performance and throughput in a range of sizes and power profiles. These cores provide customers with highly efficient, feature-rich RISC-V scalar and vector computing capabilities, tailored for optimal balance and industry-leading results.

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- ✓ High Performance 64bit Out of Order scalar engines
- ✓ Full support for RVA23 profile
- ✓ Optional vector compute engines across the portfolio



Automotive

SiFive Automotive™ solutions provide a range of high-end application and real-time processors, delivering industry-leading performance, minimal area, and low power consumption. ISO 21434, ISO 26262 ASIL B, ASIL D and split-lock certifications.

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- ✓ High performance IP for IVI, ADAS and Central Compute
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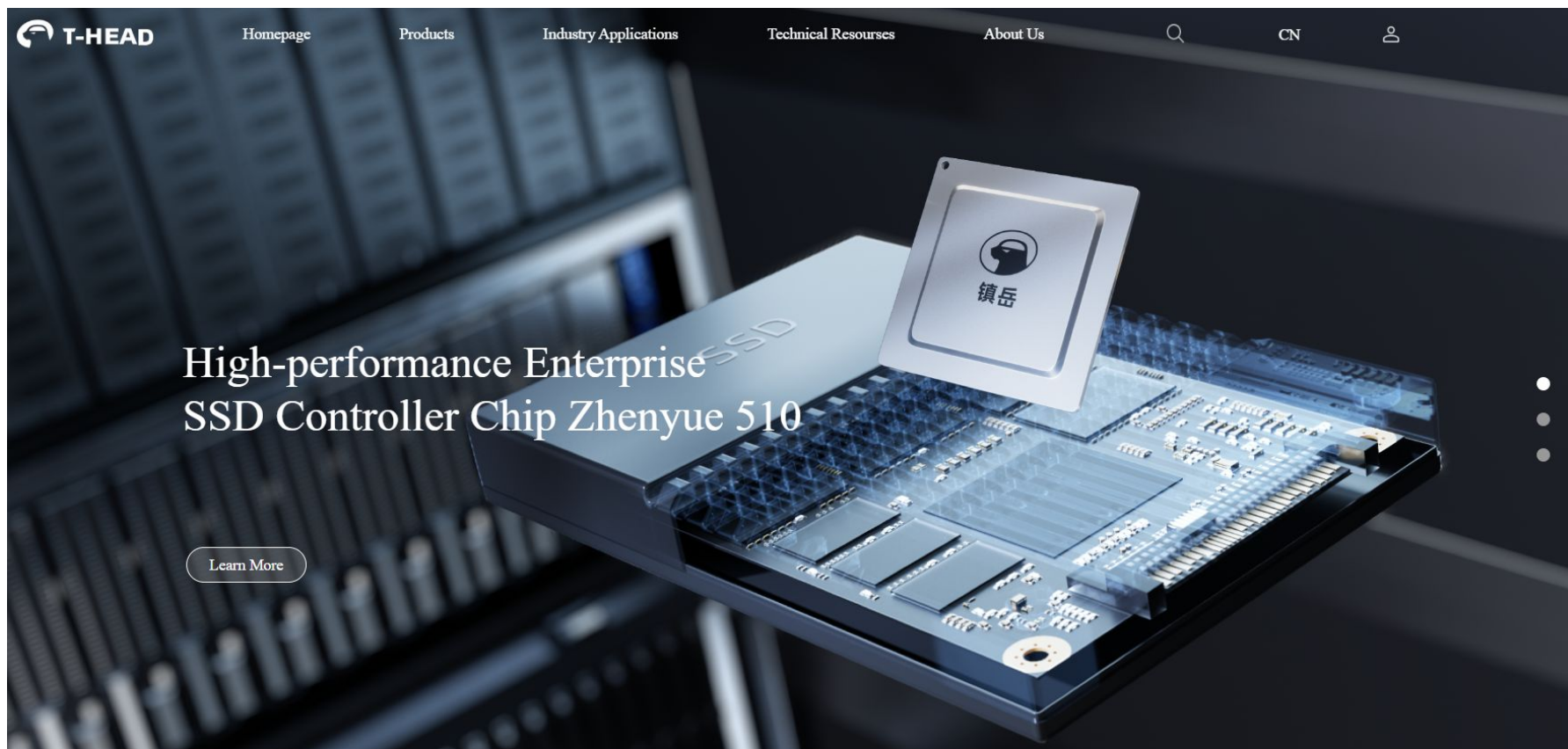
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Introduction to RIPES

RIPES is a visual computer architecture simulator for teaching RISC-V assembly and CPU design concepts.

It allows users to write RISC-V code and visualize how instructions move through the pipeline.

Key features:

- Visual datapath animation (pipeline, cache, memory).
- Step-by-step instruction execution.
- Support for single-cycle, multi-cycle, and pipelined CPUs.
- Ideal for educational use.

Writing RISC-V Assembly in RIPES

RIPES provides a text editor for entering assembly code.

Example (Add two numbers):

- `addi x1, x0, 5` # Load 5 into x1
- `addi x2, x0, 3` # Load 3 into x2
- `add x3, x1, x2` # $x3 = x1 + x2$
- `ecall` # Stop simulation

Students can view register and memory updates after each instruction.

Summary

RISC-V promotes open, modular, and simplified CPU design.

RIPES helps visualize how RISC-V CPUs execute instructions.

HDL (Verilog, VHDL, SystemVerilog) bridges design and physical implementation.

The RISC vs CISC difference lies in architectural philosophy, not programming language.

Understanding these concepts builds a foundation for advanced CPU design and optimization.